STAT2004 – Tutorial Week 11 Solutions

Exercises 2 and 3 were covered in the tutorial.

Exercise 1 (Basic properties of the multinomial distribution)

(a) Without loss of generality, we can consider the marginal distribution of X_1:

$$\begin{aligned}
P(X_1 = x_1) &= \sum_{x_2+\ldots+x_K=n-x_1} P(X_1 = x_1, X_2 = x_2, \ldots, X_k = x_K) \\
&= \sum_{x_2+x_3+\ldots+x_K=n-x_1} \frac{n!}{x_1! x_2! \cdots x_K!}\, p_1^{x_1}\, p_2^{x_2} \cdots p_K^{x_K} \\
&= \frac{n!}{x_1!}\, p_1^{x_1} \sum_{x_2+\ldots+x_K=n-x_1} \frac{1}{x_2! \cdots x_K!}\, p_2^{x_2} \cdots p_K^{x_K} \\
&= \frac{n!}{x_1!(n-x_1)!}\, p_1^{x_1}\, (1-p_1)^{n-x_1} \sum_{x_2+\ldots+x_K=n-x_1} \frac{(n-x_1)!}{x_2! \cdots x_K!} \left(\frac{p_2}{1-p_1}\right)^{x_2} \cdots \left(\frac{p_K}{1-p_1}\right)^{x_K} \\
&= \frac{n!}{x_1!(n-x_1)!}\, p_1^{x_1}\, (1-p_1)^{n-x_1} \ ,
\end{aligned}$$

because the term inside the sum is the pmf of a multinomial distribution on the remaining $K-1$ categories after the first category is removed, and so the sum = 1.

(b) Because each count $X_k \sim \text{Binomial}(n, p_k)$, its expectation is given by $E(X_k) = np_k$ and its variance is given by $\text{Var}(X_k) = np_k(1-p_k)$, respectively.

(c) Again, without loss of generality, we can consider the counts X_1 and X_2 in the first two categories. First, note that the marginal distribution of (X_1, X_2) is, by similar calculations to part (a), given by

$$P(X_1 = x_1,\ X_2 = x_2) = \frac{n!}{x_1!\, x_2!\, (n-x_1-x_2)!}\, p_1^{x_1}\, p_2^{x_2}\, (1-p_1-p_2)^{n-x_1-x_2}$$

In other words, the marginal distribution of X_1, X_2 (and the remaining $n - X_1 - X_2$ counts) reduces to a tri-nomial with probabilities p_1, p_2 (and the remaining probability $1-p_1-p_2$).

The expected value $E(X_1X_2)$ is therefore given by

$$\begin{aligned}
E(X_1X_2) &= \sum_{0\le x_1+x_2\le n} x_1\, x_2\, P(X_1 = x_1,\ X_2 = x_2) \\
&= \sum_{0\le x_1+x_2\le n} x_1\, x_2\, \frac{n!}{x_1!\, x_2!\, (n-x_1-x_2)!}\, p_1^{x_1}\, p_2^{x_2}\, (1-p_1-p_2)^{n-x_1-x_2} \\
&= \sum_{0\le (x_1-1)+(x_2-1)\le n-2} \frac{n!}{(x_1-1)!\, (x_2-1)!\, (n-x_1-x_2)!}\, p_1^{x_1}\, p_2^{x_2}\, (1-p_1-p_2)^{n-x_1-x_2}
\end{aligned}$$

To handle the term in the sum, write $\tilde{n} = n - 2$, $\tilde{x}_1 = x_1 - 1$ and $\tilde{x}_2 = x_2 - 1$, so that

$$\begin{aligned} n! &= n(n-1)(n-2)! &&= n(n-1)\tilde{n}! \\ (n - x_1 - x_2)! &= [n - 2 - (x_1 - 1) - (x_2 - 1)]! &&= (\tilde{n} - \tilde{x}_1 - \tilde{x}_2)! \\ p_1^{x_1} &= p_1 \times p_1^{x_1 - 1} &&= p_1 \times p_1^{\tilde{x}_1} \\ p_2^{x_2} &= p_2 \times p_2^{x_2 - 1} &&= p_2 \times p_2^{\tilde{x}_2} \quad \text{and} \\ (1 - p_1 - p_2)^{n - x_1 - x_2} &= (1 - p_1 - p_2)^{n - 2 - (x_1 - 1) - (x_2 - 2)} &&= (1 - p_1 - p_2)^{\tilde{n} - \tilde{x}_1 - \tilde{x}_2} \end{aligned}$$

Then, the expected value $E(X_1 X_2)$ can be written as

$$\begin{aligned} E(X_1 X_2) &= n(n-1)\, p_1\, p_2 \sum_{0 \le \tilde{x}_1 + \tilde{x}_2 \le \tilde{n}} \frac{\tilde{n}!}{\tilde{x}_1!\, \tilde{x}_2!\, (\tilde{n} - \tilde{x}_1 - \tilde{x}_2)!}\, p_1^{\tilde{x}_1}\, p_2^{\tilde{x}_2}\, (1 - p_1 - p_2)^{\tilde{n} - \tilde{x}_1 - \tilde{x}_2} \\ &= n(n-1)\, p_1\, p_2 \ , \end{aligned}$$

because the term in the sum is the pmf of a Multinomial$(\tilde{n}, p_1, p_2, \ldots, p_K)$ distribution, and so the sum = 1.

Thus, the covariance between X_1 and X_2 is given by

$$\begin{aligned} \mathrm{Cov}(X_1, X_2) &= E(X_1 X_2) - E(X_1) E(X_2) \\ &= n(n-1)\, p_1\, p_2 - (n\, p_1)(n\, p_2) \\ &= -n\, p_1\, p_2 \ . \end{aligned}$$

(d) For $K = 2$ categories, the covariance of X_1 and X_2 is given by part (c) as $-np_1(1-p_1)$ and both variances are given by part (b) as $np_1(1-p_1)$. Hence, the correlation between is

$$\mathrm{corr} = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\mathrm{Var}(X_2)}} = -1 \ ,$$

that is, they are perfectly negatively correlated. Of course, we didn't need to use any formulae for this, because when there are only two categories then the count in the first category completely determines the count in the second category, and vice versa.

Exercise 2 (Goodness-of-fit test for a die)

	1	2	3	4	5	6
counts	17	15	12	18	13	25

We are interested in testing whether this die is fair or loaded.

(a) The null hypothesis is that the die is fair, i.e., $H_0 : \boldsymbol{p} = \boldsymbol{p}^* = (1/6, 1/6, 1/6, 1/6, 1/6, 1/6)^\top$. The alternative hypothesis is that the die is loaded, i.e., $H_1 : \boldsymbol{p} \neq \boldsymbol{p}^*$, or equivalently, H_1 : not all $p_i = 1/6$, or equivalently, H_1 : at least one $p_i \neq 1/6$.

(b) If the null hypothesis is indeed true, then we would expect to see $100/6 = 16.67$ counts for each side of this die.

(c) The generalized likelihood ratio test statistic for these data is

$$\Lambda = \frac{L(\boldsymbol{p}^*|x)}{L(\hat{\boldsymbol{p}}|x)} = \frac{(\frac{1}{6})^{100}}{(\frac{17}{100})^{17}(\frac{15}{100})^{15}(\frac{12}{100})^{12}(\frac{18}{100})^{18}(\frac{13}{100})^{13}(\frac{25}{100})^{25}} = 0.045 \ .$$

The transformation of the GLRT is therefore

$$-2\log\Lambda = -2\log(0.045) = 6.212 \ .$$

Pearson's χ^2 test statistic is

$$\begin{aligned} \chi^2 &= \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i} \\ &= \frac{(17-16.67)^2}{16.67} + \frac{(15-16.67)^2}{16.67} + \frac{(12-16.67)^2}{16.67} \\ &\quad + \frac{(18-16.67)^2}{16.67} + \frac{(13-16.67)^2}{16.67} + \frac{(25-16.67)^2}{16.67} \\ &= 6.559 \ . \end{aligned}$$

Pearson's χ^2 statistic and the exact GLRT are, again, very similar numerically.

(d) Under the null hypothesis, Pearson's χ^2 statistic has an approximate χ^2_5 distribution.

(e) The p-value is $P(\chi^2_5 \geq 6.559) = 0.256$. There is therefore no evidence that the die is loaded and we can retain the claim that the die is fair

In Tutorial 8 Exercise 3, recall that we found *moderately strong* evidence that the proportion of "sixes" p_6 is not equal to 1/6 for this die (with an associated p-value of 0.043).

(f) The p-value from Tutorial 9 Exercise 4 was obtained by testing the "most unusual" category that we happened to see from data (i.e., the "sixes"), whereas the p-value from part (e) above takes into account the outcome of all six sides. If we look at the data first and then choose the "most unusual" subset of outcomes to test, then we can almost always be able to force a "significant" result. This is called *data snooping*.

(g) Given that there are six sides, and any one of these six sides could have been the "most unusual" outcome, we can apply a quick-and-dirty Bonferroni correction by multiplying the p-value of 0.043 by 6 = 0.26. After adjusting for multiplicity, we again find no evidence that the die is loaded.

Exercise 3 (Are suicide rates constant throughout the year?[1])

(a) The overall suicide rate was $23480/365 = 64.329$ per day

(b) If the suicide rate is indeed constant throughout the year, then the expected number of suicides in each month will be $64.329\times$ number of days in the month. For January, this would be $64.329 \times 31 = 1994.19$, and so on for the other 11 months. We can write the expected counts next to the observed counts in the table:

Month	Number of Suicides	Expected Number	$\frac{O_i - E_i}{\sqrt{E_i}}$	Days/Month
Jan.	1867	1994.19	-2.85	31
Feb.	1789	1801.21	-0.29	28
Mar.	1944	1994.19	-1.12	31
Apr.	2094	1929.86	3.74	30
May.	2097	1994.19	2.30	31
Jun.	1981	1929.86	1.16	30
Jul.	1887	1994.19	-2.40	31
Aug.	2024	1994.19	0.67	31
Sep.	1928	1929.86	-0.04	30
Oct.	2032	1994.19	0.85	31
Nov.	1978	1929.86	1.10	30
Dec.	1859	1994.19	-3.03	31
Total	23480	23480.00		365

(c) Pearson's χ^2 statistic computed on these data is

$$\begin{aligned}\sum_{\text{cells}i} \frac{(O_i - E_i)^2}{E_i} &= \frac{(1867 - 1994.19)^2}{1994.19} + \frac{(1789 - 1801.21)^2}{1801.21} + \ldots + \frac{(1859 - 1994.19)^2}{1994.19} \\ &= 47.37\end{aligned}$$

The null distribution of this test statistic is χ^2 with 11 degrees of freedom. This is because the null hypothesis completely specifies the incidence rate p_i of each month as being directly proportional to the number of days in each month.

(d) The p-value here is $P\left(\chi^2_{11} \geq 47.37\right) = 1.85 \times 10^{-6}$ from `R`. If the rate of suicides is indeed uniform throughout the year, then observing counts with such large deviations between months (or even larger) would occur by chance less than $0.000002 = 0.0002\%$ of the time (i.e., almost impossible). There is extremely strong evidence that the rate of suicide is not constant throughout the year.

(e) The signs and magnitudes of the individual *relative deviations* $(\mathrm{O}_i - \mathrm{E}_i)/\sqrt{\mathrm{E}_i}$ for each month i are given in the table above. We see that the suicide rate appeared to be lowest in the US winter months, and highest during April/May, the beginning of the third Financial quarter. We can speculate what was driving this trend, but we cannot show causality with these data.

[1] Adapted from Exercise 9.28 from *Mathematical Statistics & Data Analysis* (Rice, 1995).

Exercise 4 (Assessing authorship via word usage[2])

(a) The null hypothesis is that the relative frequency of usage of these six words is the same across all three Austen works. The alternative hypothesis is that at least one of these three works uses these six words with a different relative frequency compared to the others.

In the three Austen works, the word "a" was used 434 times out of a total of 1,017 words. So if the relative frequency of word usage is the same across all three works, then the expected count of "a" in *Sense and Sensibility"* would be $375 \times 434/1017 = 160.03$. Repeating these calculations for all cells in the table involving only Austen's works gives the following set of expected values (in parentheses):

Word	*Sense and Sensibility*	*Emma*	*Sanditon (Austen)*	Total
a	147 (160.03)	186 (187.77)	101 (86.20)	434
an	25 (22.86)	26 (26.82)	11 (12.31)	62
this	32 (31.71)	39 (37.21)	15 (17.08)	86
that	94 (87.02)	105 (102.10)	37 (46.88)	236
with	59 (59.37)	74 (69.66)	28 (31.98)	161
without	18 (14.01)	10 (16.44)	10 (7.55)	38
Total	375	440	202	1017

Pearson's χ^2 statistic is calculated as

$$\begin{aligned}
\chi^2 &= \sum_{\text{cells } ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \\
&= \frac{(147 - 160.03)^2}{160.03} + \frac{(186 - 187.77)^2}{187.77} + \ldots + \frac{(10 - 7.55)^2}{7.55} \\
&= 12.271 \quad \text{on } (6-1) \times (3-1) = 10 \text{ degrees-of-freedom.}
\end{aligned}$$

The p-value for testing homogeneity is given by $P(\chi^2_{10} \geq 12.271) = 0.267$. We can conclude that Austen's usage of these six words were consistent over these three works.

(b) In light of our findings from part (a), we can pool together the word counts from Austen's three works. Carrying out another test of homogeneity between Austen's pooled word count and the imitator's, we find the following observed and expected counts:

Word	Austen	Imitator
a	434 (433.5)	83 (83.5)
an	62 (76.3)	29 (14.7)
this	86 (84.7)	15 (16.3)
that	236 (216.3)	22 (41.7)
with	161 (171.0)	43 (33.0)
without	38 (35.2)	4 (6.8)
Total	1017	196

Pearson's χ^2 statistic is calculated as

$$\begin{aligned}
\chi^2 &= \sum_{\text{cells } ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \\
&= \frac{(434 - 433.5)^2}{433.5} + \frac{(83 - 83.5)^2}{83.5} + \ldots + \frac{(4 - 6.8)^2}{6.8} \\
&= 32.81 \quad \text{on } (6-1) \times (2-1) = 5 \text{ degrees-of-freedom.}
\end{aligned}$$

The p-value for testing homogeneity is given by $P(\chi^2_5 \geq 32.81) = 0.00041 = 0.04\%$. There is very strong evidence that the imitator's word usage was not consistent with Austen's word usage.

Of course, the manual calculations here are incredibly tedious (although not hard). We can automate this process in `R` using following commands:

```
#############################################################################
# Part (a)
# input all of Austen's counts as a single vector
Austen.counts = c(147,186,101,
                  25, 26, 11,
                  32, 39, 15,
                  94,105, 37,
                  59, 74, 28,
                  18, 10, 10)

# convert Austen's counts to a 6x3 table
Austen.table = matrix(Austen.counts, nrow=6, byrow=T)

# perform Pearson's chi..sq test of homogeneity
# and save it as "test1"
test1 = chisq.test(Austen.table)

# examine the results of "test1"
test1

# extract the expected values from "test1"
test1$expected

#############################################################################
# Part (b)
# calculate the row totals from Austen's table of counts:
# in the command below, we "apply" the function "sum" to
# to the rows (i.e., dimension "1") of Austens.table
pooled.counts = apply(Austen.table, 1, sum)
```

```
# input all of the imitators counts as a vector
Imitator.counts = c(83,29,15,22,43,4)

# combine Austen's pooled word counts with Imitator's into a 6x2 table
combined.table = matrix(c(pooled.counts,Imitator.counts),nrow=6,byrow=F)

# perform Pearson's chi..sq test of homogeneity
# and save it as "test1"
test2 = chisq.test(combined.table)

# examine the results of "test1"
test2

# look at the table of relative deviations (O_i - E_i)/sqrt(E_i):
test2$residuals

            [,1]        [,2]
[1,]  0.02585693 -0.05889921
[2,] -1.63667544  3.72816398
[3,]  0.14342992 -0.32671736
[4,]  1.33865901 -3.04931581
[5,] -0.76747283  1.74821745
[6,]  0.46957090 -1.06963011
```

(c) Looking at the *relative deviations* $(O_i - E_i)/\sqrt{E_i}$ from the output above we see that compared to Jane Austen, the imitator overused "an" and underused "that" the most.

[2] Adapted from Section 13.3 of *Mathematical Statistics & Data Analysis* (Rice, 1995).